



Introduction

Department of Computer Engineering

Sharif University of Technology

Hamid R. Rabiee rabiee@sharif.edu

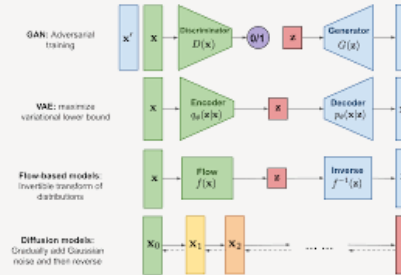
Maryam Ramezani maryam.ramezani@sharif.edu



Course Description

How is our course?

Machine Learning Mathematics





Machine Learning Mathematics

01

Linear Algebra

Provides the foundation for manipulating data in high-dimensional spaces, essential for vector operations in machine learning.

02

Statistics & Probability

Offers tools to model uncertainty and make inferences about data, forming the backbone of many machine learning algorithms.

03

Optimization

Focuses on finding the best parameters for a model by minimizing or maximizing an objective function.

04

Information Theory

Analyzes data compression and transmission, helping quantify uncertainty and information gain in machine learning models.

05

Signal Processing

Involves analyzing and transforming data signals, important for handling time-series data and feature extraction in machine learning.

06

Graph Theory

Studies networks of nodes and edges, crucial for understanding relationships in data and complex models like Graph Neural Networks.

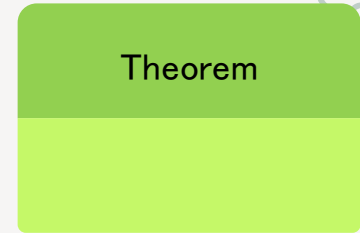
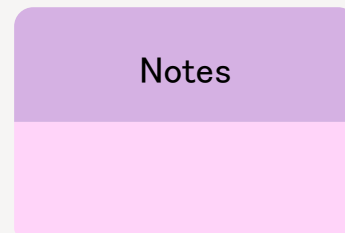
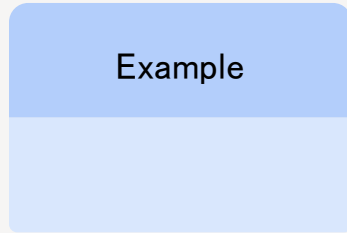
Course Material

<https://sut-ce-courses.github.io/linearalgebra> (jabrekh.ir)

https://quera.org/course/add_to_course/course/29169/ (password: Fall2026LA11)

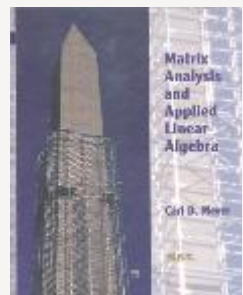
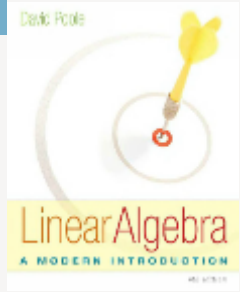
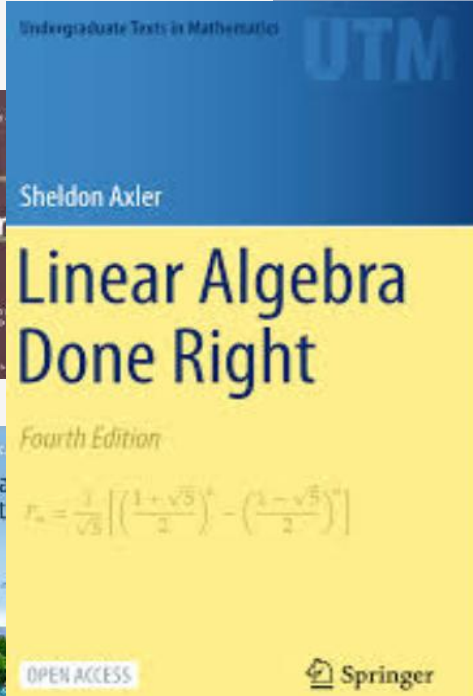
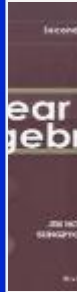
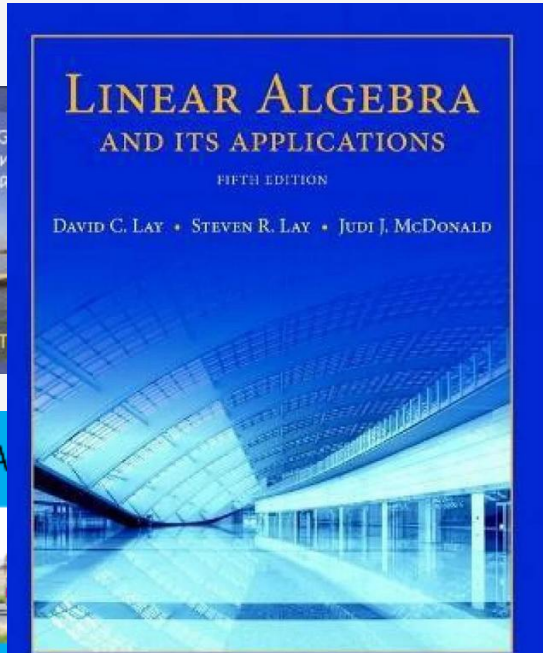
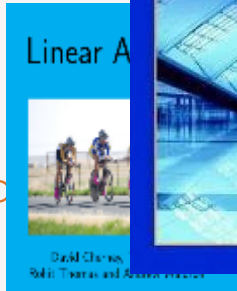
Course Method

- Slides



- Writing on board
- Geometric Interpretation and Intuition
- Notebooks

Course References



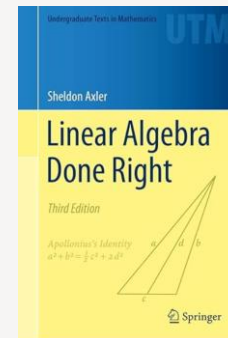
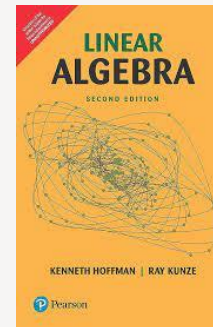
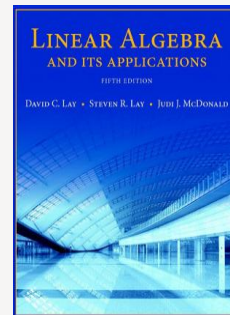
Reach Us!

- Instructors:
 - Hamid R. Rabiee: (CE804) rabiee@sharif.edu
 - Maryam Ramezani: (CE821) maryam.ramezani@sharif.edu
- Course notes, homework and solutions, handouts, and other useful resources are available on the course page:
 - [https://quera.org/course/add_to_course/course/29169/\(password: Fall2026LA11\)](https://quera.org/course/add_to_course/course/29169/(password: Fall2026LA11))
- Class: Hall 4 (Sunday & Tuesday: 10:30-12:00)
 - <https://vc.sharif.edu/ch/maryam.ramezani> & <https://vc.sharif.edu/ch/rabiee>
- Lead TA:
 - Amirmahdi Vaziri (Amirmahdi.vaziri2005@gmail.com, Telegram ID : [@kiavzr](https://t.me/@kiavzr))
 - Telegram Group: <https://t.me/+xBhtmlEi32o0NWE0>

Resources

- Textbooks:
 - Sheldon Axler, *Linear Algebra Done Right*, fourth edition, 2023
 - Kenneth Hoffman and Ray A. Kunze. *Linear Algebra*. PHI Learning, 2004.
 - Gilbert Strang. *Introduction to Linear Algebra*. Wellesley-Cambridge Press, 2016.
 - David C. Lay, Steven R. Lay, and Judi J. McDonald. *Linear Algebra and Its Applications*. Pearson, 2016.

+Other textbooks and course materials.



Structure of the Course

Lectures

Goal: To introduce concepts in linear algebra, and motivate their use and importance.

Note: We try to cover useful materials in class, but we recommend you reading more!

Exams

Quiz: 5 with dropping lowest

Midterm: 1

Final: 1

Assignments

Purpose: To give you a chance to exercise your mind, and to solidify the concepts introduced to you in class.

Structure: Seven theoretical assignments, and Two linear algebra practical assignments.

Importance: Not important unless you want to learn the material and get a good grade.

Number of Exercises: Written: 7 Programming: 1

Grading Policy

Quizzes

3.6 points (5 quizzes each 0.9 points; the lowest score will be dropped)

Theoretical Assignments

3.5 points (each 0.5 points)

Programming Assignment

0.9 points (1 Assignment)

Midterm

6 points (Lectures part 1,2,3,4)

Final

6 points (Lectures part 5,6,7)

Total: **20** Points

Lecture Notes

- Lecture slide will be uploaded.
- Many times we will write on board, in real-time, during lecture to prove a theory or answer a question or add some additional explanations. It will be your responsibility to take notes.
- Slides links will be provided on site.

Syllabus

1405/07/05	S01	1	Introduction
1405/07/07	S02		Elementary Row Operations, and Linear Equations
1405/07/12	S03		Row Reduction, Echelon Forms and LU Factorization
1405/07/14	S04		Vector Space
			Subspace

1405/07/19	S05	2	Linear Independence
1405/07/21	S06		
1405/07/26	S07		Bases, Dimension
1405/07/28	S08		
1405/08/03	S09		Matrix Rank

Syllabus

1405/08/05	S10	3	Linear Transformation
1405/08/10	S11		
1405/08/12	S12		Change Basis
1405/08/17	S13		Inverse

1405/08/19	S14	4	Inner Product Space and Norm2
1405/08/24	S15		
1405/08/26	S16		Orthogonality (Gram–Schmidt, QR , etc.)

Syllabus

1405/09/01	S17	5	Determinant
1405/09/03	S18		
1405/09/08	S19		Eigenvectors and Eigenvalues
1405/09/10	S20		Similarity & Diagonalization

1405/09/15	S21	6	Symmetric Matrices and Quadratic Forms
1405/09/17	S22		Matrix Factorization
1405/09/19	Midterm Exam		
1405/09/22	S23		Singular Values and Singular Vectors
1405/09/24	S24		SVD
1405/09/29	S25		

Syllabus

1405/10/01	S26	7	Norm Space
1405/10/06	S27		Vector and Matrix Derivatives
1405/10/08	S28		
1405/10/13	S29		Least squares
1405/11/06	Final Exam		

Assignments

- Theoretical assignments are released at the end of each part. 3 questions of each assignment are homework. Each has 0.5 points
- Programming Assignments: 0.9 points

Assignments

Assignment	Release	Deadline
T1	1405/07/19	1405/07/28
T2	1405/08/05	1405/08/17
T3	1405/08/19	1405/09/01
T4	1405/09/01	1405/09/10
T5	1405/09/15	1405/09/29
T6	1405/10/01	1405/10/13
T7	1405/10/15	1405/10/25

Exams

Quizzes will be held as the following table on 12:00 for half an hour and will be graded in Grade Scope.

Exam	Date
Quiz 1	3 Aban
Quiz 2	19 Aban
Quiz 3	3 Azar
Midterm	19 Azar
Quiz 4	1 Dey
Quiz 5	15 Dey
Final	6 Bahman

TA Classes

Date	Title
21 Mehr	Quiz1
10 Aban	Quiz2
26 Aban	Quiz3
17 Azar	Midterm
24 Azar	Quiz4
6 Dey	Quiz5
22 Dey	Final



Introduction

Lets think about a question!

01

Vector

Basic Concept

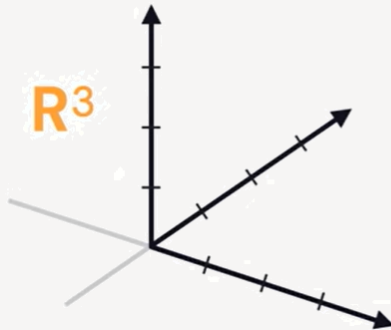


Data Representations (Linear Algebra)

- How can we represent data (images, text, user preferences, etc.) in a way that computers can understand?
- Organize information into a vector!
 - A vector is a 1-dimensional array of numbers.
 - It has both a magnitude (length) and a direction
- The totality of a vectors with n entries is an n-dimensional vector space.

$$V = \begin{matrix} \nearrow \\ = \end{matrix} \begin{bmatrix} -3 \\ 0.7 \\ 2 \end{bmatrix}$$

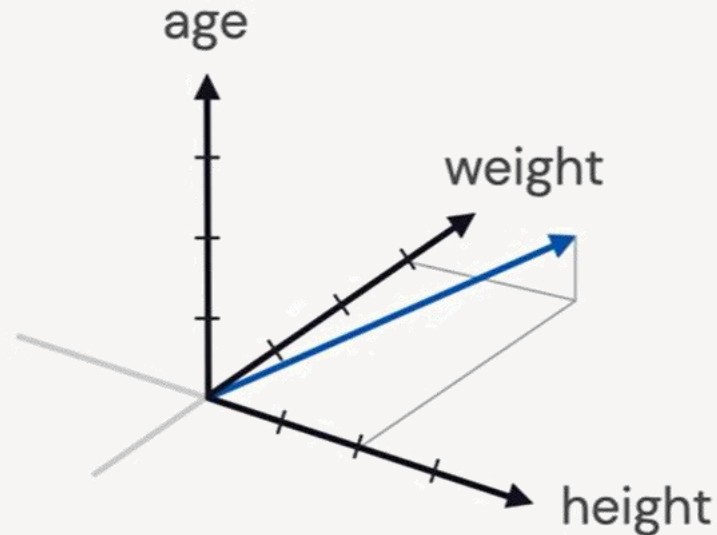
“3-dimensional space” consists
of all vectors with 3 entries:



Data Representations (Machine Learning)

- A **feature vector** is a vector whose entries represent the “features” of an object.
- The vector space containing them is called **feature space**.

$$P = \begin{bmatrix} 64 \\ 131 \\ 24 \end{bmatrix} \begin{array}{l} \text{height} \\ \text{weight} \\ \text{age} \end{array}$$





02

Equation with Matrix and Vector Format



House Pricing Example



Price Problem



\$ 70'000



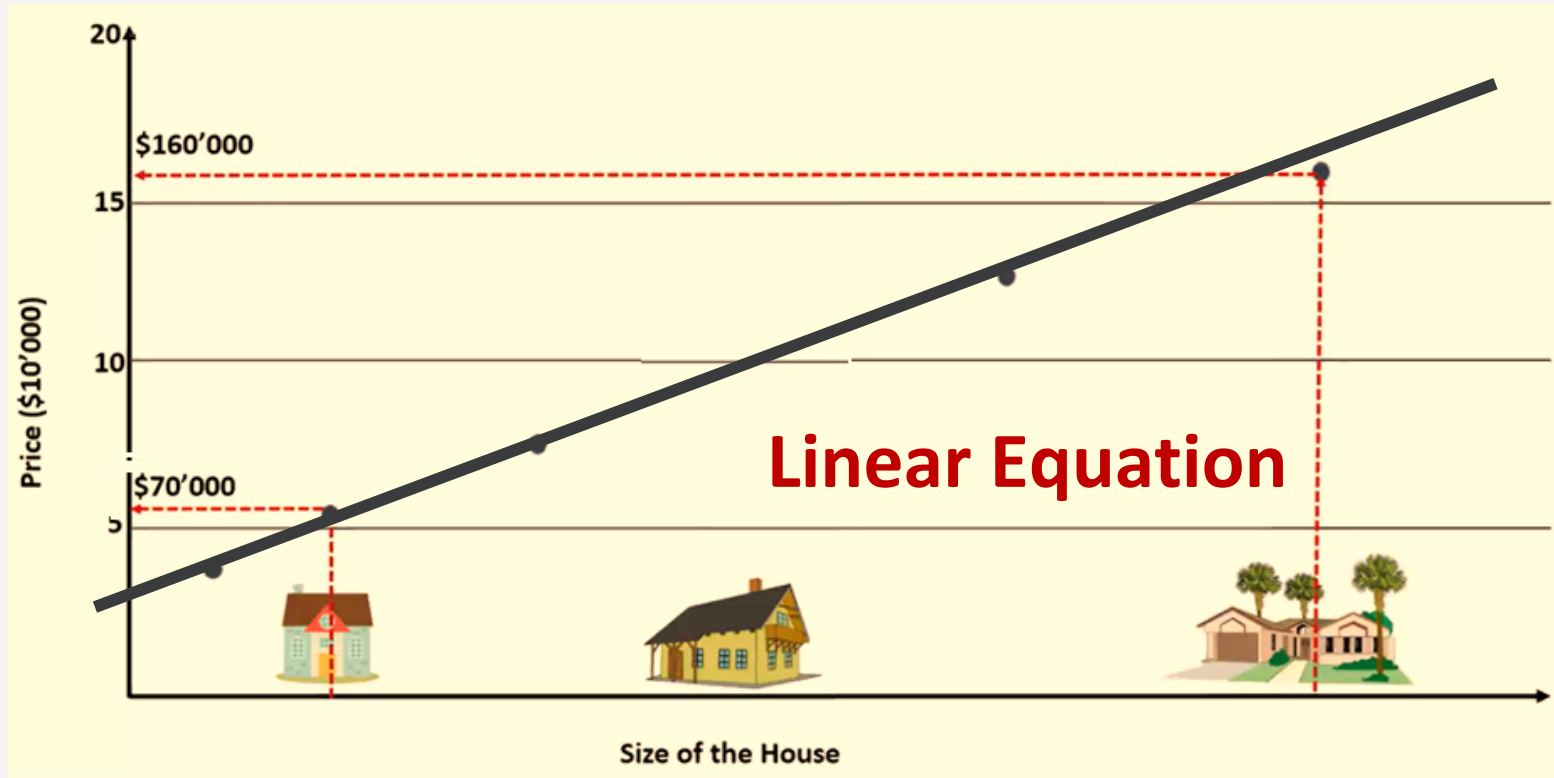
?



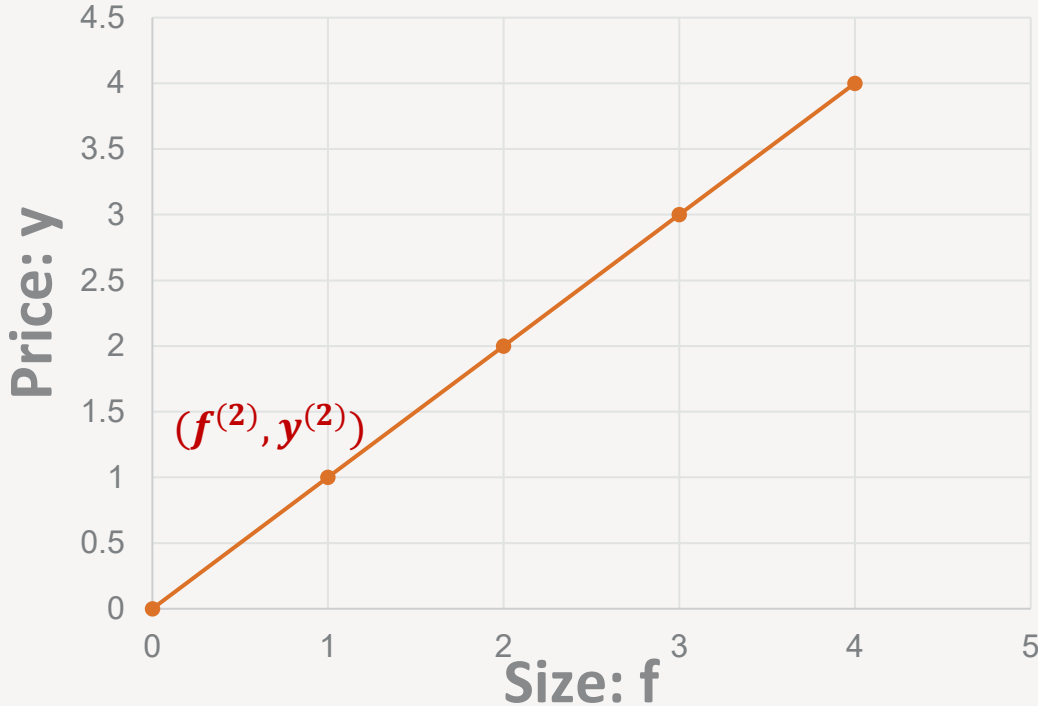
\$ 160'000



Price Problem



Linear Equation without offset



$$y = af$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$



Linear Equation with offset



$$y = af + c$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

House Features

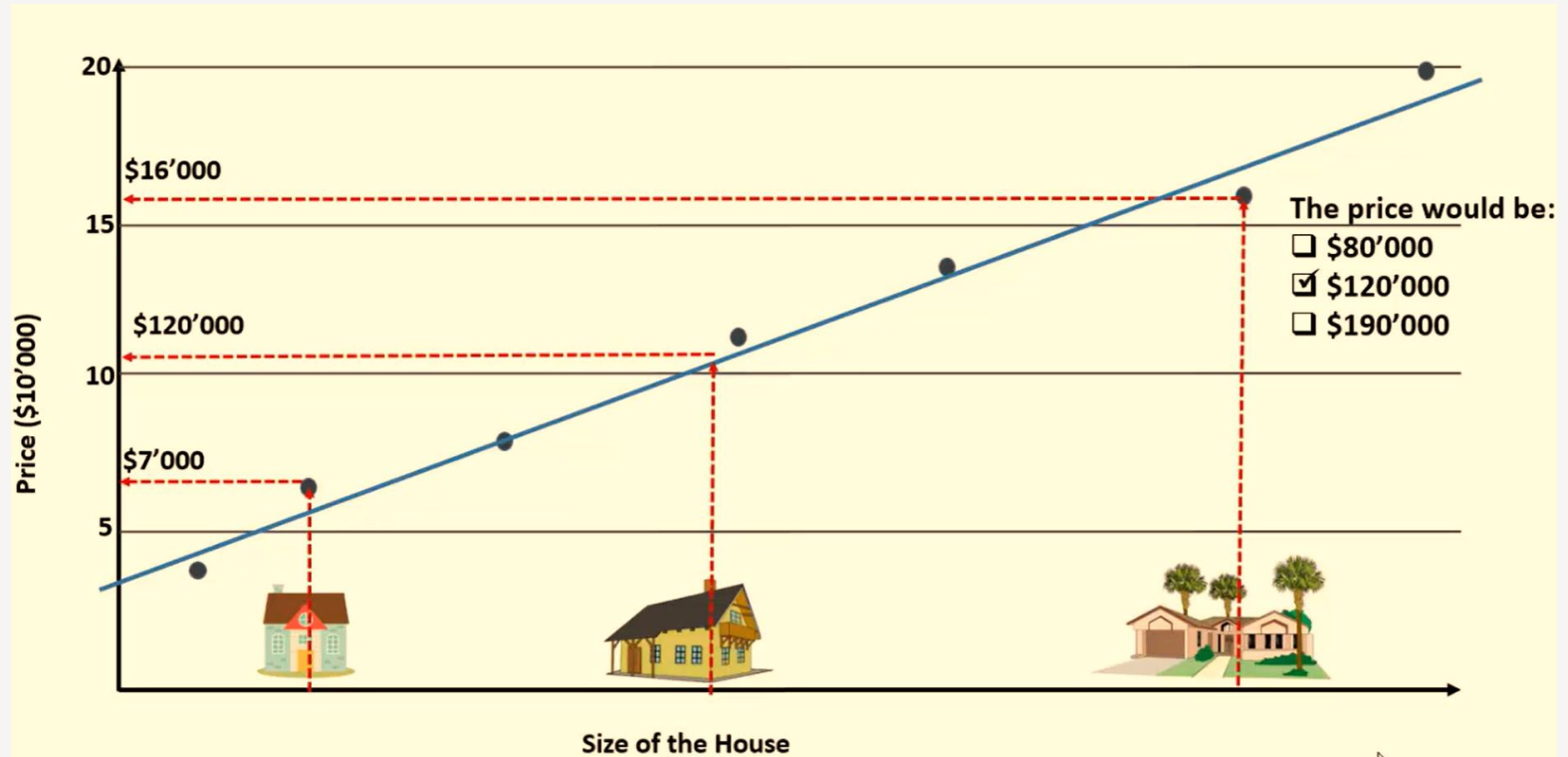
- #Room
- Size
- #Bedroom
- Age
- Address features: Street, Alley, ...
- Size of part1, part2, part3, part4
- Floors
- #Bathrooms

N number of training data with M features:

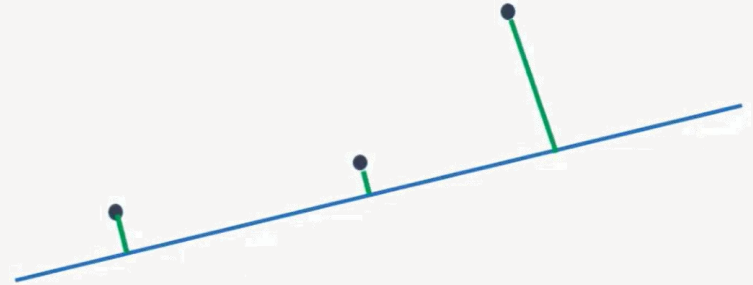
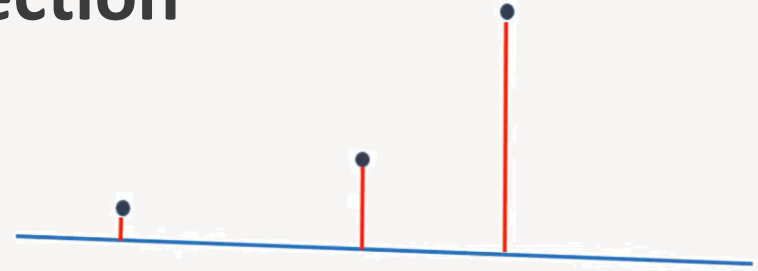
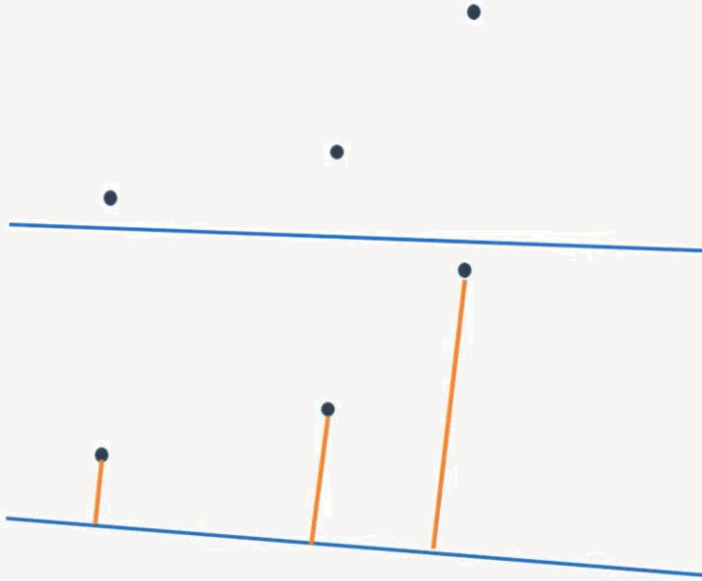
$$A_{N \times M} x_{M \times 1} = b_{N \times 1}$$

Linear Equation

Least Squares Error Correction



Least Squares Error Correction



Error 1: 

Error 2: 

Error 3: 

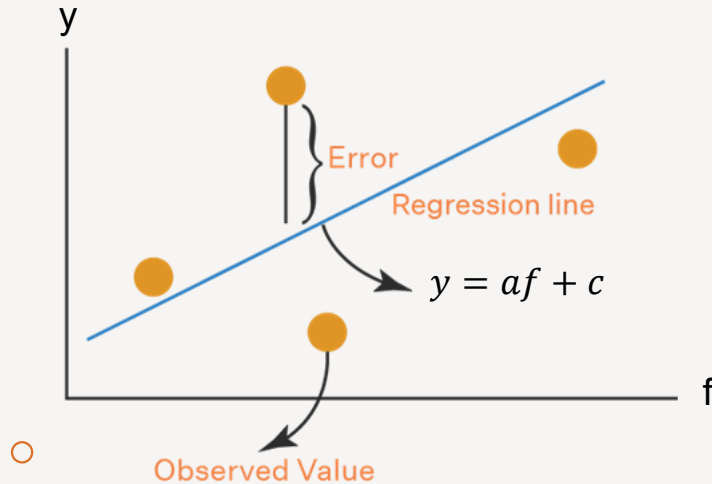
Least Squares

Least Square Method

- Objective for “m” features:

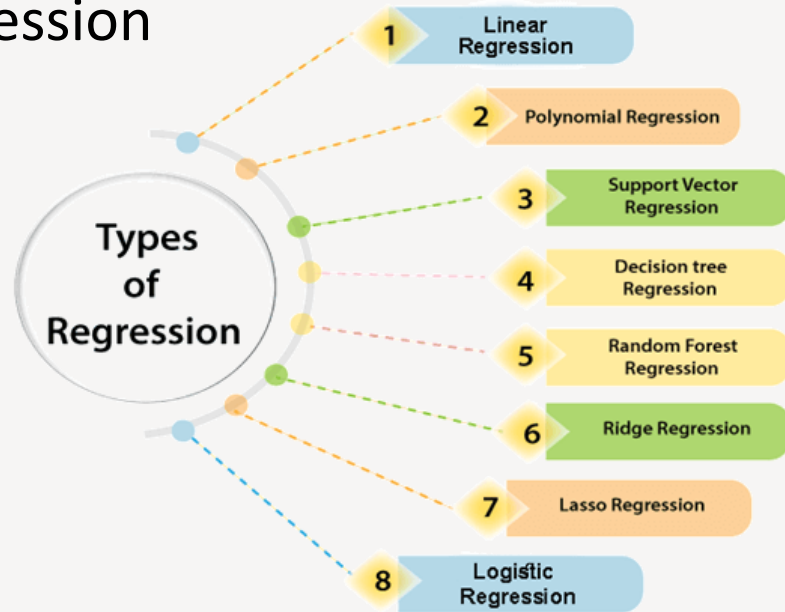
$$\hat{y} = a_1 f_1 + a_2 f_2 + \cdots + a_m f_m + c$$

$$\min ||y - \hat{y}||$$



Linear Algebra and Machine Learning Application

- $Ax = b \rightarrow x = A^{-1}b$ Inverse of Matrix/Pseudo Inverse
- Regression



02

Road Map

Step by Step

